

Python Logic and Modularity – Analysis Questions

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Analysis Questions

1. How do you think the Python program you wrote could be made more efficient?
2. How is the context of each tool (irrigation check vs. escape velocity) important to how you designed its inputs, validation, and outputs?
3. What are the benefits of organizing two unrelated tools into one program with a menu, compared to writing two separate scripts? What are the drawbacks?
 - One aspect could be using a single function that does the conversion from the yes/no answers for pressure and maintenance requirements to the True, False, or warning. Otherwise, the tools seem reasonably efficient for their complexity.
 - The irrigation check required both numerical and boolean values to check its validity. This involved turning the inputs into corresponding floats and booleans (from yes or no) while the validation itself is made up of multiple if-elif-else statements to go down the conditions in the proper priority. The outputs were simple string statements and were therefore just printed out. The escape velocity calculator on the other hand only required float value inputs. The processing involved simple mathematical operations using Python's math library. The output for the calculator also needed to concatenate and round the outputs with strings.
 - When placed into one program, both tools are available while only running one script. Having two unrelated scripts has some drawbacks in that the code itself could get too cluttered or large with many tools in one file. Additionally, if only one tool is required, the other script is still defined, taking up minor resources in this case but still unnecessary and could compound with more tools or if the script is ran many times.